Outbreak: Riverton

Scientific Response Director · Riverton

10 STAGES · 40 STOPS · GRADE 6 · OUTBREAK_RIVERTON_MS

THE OPENING CARD

Three hospitals in one river city have reported the same strange illness within a day of each other. The ambulances are still arriving. Seven patients, four days, and the same four symptoms every time. You are the science lead for the city, which means what gets measured is your call, and the city acts on whatever comes back. Sofia Morales will not yet say whether seven is a lot, because nobody wrote down what a normal week looks like. An outbreak can only be stopped while it is small, and this one has had four days.

STAGE 1 OF 10

The First Cluster

BEFORE THE STAGE — GREET

Six leads, three shifts, and a fence between them

The ward side and the laboratory side have hardly met. By Friday one will have to trust the other's number. Put a name to every lead before the shift changes, because a phone call at three in the morning goes better between people who have met.

PLAN CARD 70W

Day one. Seven patients, three hospitals, four days, and the same four symptoms in every one of them. Sofia Morales has the printout, and a reporter downstairs already has the number. Today you decide whether this is an outbreak at all, and agree what counts as a case. Seven sounds like a lot. Morales will not say whether it is until she knows what a normal week here looks like.

BRIEFING 34W

Three Riverton hospitals report an unusual combination of fever, dry cough, severe headache, and profound fatigue. The player must decide whether the pattern is a true outbreak signal before scarce emergency resources are committed.

WHAT YOU DECIDE 14W

Open an investigation using a defensible provisional case definition and an explicit comparison baseline.

1.1 Signal or noise?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · CHOICE

SCENE 30W

Seven patients across three hospitals in four days, with the same four symptoms. Dr. Sofia Morales has the printouts, and the press office has asked twice whether to say anything.

GUIDE 64W

Four options, and one of them says nothing can be known at all. Ask of each what it would take to be true. A count on its own is not a signal. It becomes one when you have something to compare it with, and this city normally sees about two a year. Two of the seven were found another way, outside the alert rule.

ASK 17W

The city normally sees about two of these a year. How many were there in four days?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a rate counts cases against how many people there are and how long it took · the system only sees the patients who reached a hospital

VERDICT 78W

A count on its own is not a signal. What makes it one is a comparison with what was expected, written down beforehand. This city logs about two of these a year. Seven in four days is not a busy week. The quiet reading matters most: two of the seven were found by hand, outside the alert that found the other five. If every case had come through one alert, the cluster could have belonged to the alert.

TAKEAWAY 13W

A count becomes a signal only when something says what was expected instead.

1.2 What counts as a case

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 36W

Three hospitals start counting at six in the morning. Each has written its own idea of what counts. Dr. Lena Ortiz, who leads the ward side, wants one on every desk before the night shift leaves.

GUIDE 61W

All four steps will happen. Ask of each what it lets you check. A rule for what counts is a tool, and it can be wrong in two directions. Too narrow and you miss the spread. Too wide and it fills up with ordinary coughs. So it gets tried on known cases and on clear non-cases, and those catch opposite mistakes.

ASK 13W

Put the four steps in the order each one can be checked in.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a graph of cases only means something if everybody counted the same thing · a definition can be too narrow or too wide, and both are wrong

VERDICT 73W

A definition of what counts is an instrument, and it can be wrong in two directions. Too narrow and most of the spread is invisible. Too wide and the count fills up with ordinary autumn coughs. That is why it gets tested twice before it goes out. Against patients everybody agrees are cases, which catches one that is too narrow. Against people who obviously are not, which catches one that is too wide.

TAKEAWAY 14W

A definition turns a worry into something two people can count the same way.

1.3 Spend the first afternoon

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · CHOICE

SCENE 24W

Morales can start one thing this afternoon. Four are ready to go, and tomorrow's hospital numbers will land before any second one can start.

GUIDE 58W

All four options are real facts about a case. Ask of each whether three hospitals could add their numbers together without agreeing on it. Two hospitals counting different things produce a total that means nothing. The name, the place and the time are all worth having, and none of them changes what the number is a count of.

ASK 18W

Three hospitals are sending numbers tomorrow. What has to be the same before they can be added together?

BEHIND THE BUTTON 169W

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TAKES AS READ

three hospitals can only be added together if they counted the same way

VERDICT 71W

Three hospitals are about to send in three sets of numbers. They can only be added together if the same rule made them. A shared list, with the same dates and the same town figures, is what does that. Testing the sewage gives a separate signal about the whole town, and it cannot say who is ill. Asking experts what they think has nothing underneath it. A warning adds no evidence.

TAKEAWAY 13W

The best first measurement is the one that makes everything measured later comparable.

END OF THE STAGE 20W

Use baseline variation, case definition, controls, correlation versus coincidence, measurement bias as an evidence chain rather than as isolated facts.

STAGE 2 OF 10

What Kind of Agent?

BEFORE THE STAGE — TRIAL

Walk the hospital before you are needed on it

It is week three, and the hospital has been rebuilt around the outbreak. Tents in the courtyards, a wash tunnel across the main path, labs in containers. Walk it once, because a route you have not walked is one you will take in dirty gloves.

PLAN CARD 75W

Day two. Pathology has four results from the same samples and will not put a name to any of them. The pictures down the microscope are the most convincing thing anybody has seen all week. That is exactly why Maya Chen does not trust them yet. Today you decide what broad kind of thing this is. Everything after today is chosen off that call: what to grow it in, and what tomorrow's test hunts for.

BRIEFING 33W

Initial samples contain biological material, but the microscope and the routine tests disagree. The team must distinguish a cellular pathogen from a virus or a noninfectious cause without relying on one ambiguous image.

WHAT YOU DECIDE 11W

Classify the broad agent category and identify the next discriminating test.

2.1 What kind of germ fits everything?

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · CHOICE

SCENE 33W

Pathology has four results from the same samples and still cannot name what this is. The samples are waiting under the hood. Ortiz needs a broad answer before the overnight run is booked.

GUIDE 53W

Four options, and three results on the bench are negatives. Nothing cell-shaped on the slide. No bacterial marker. No fungal wall. Ask of each option how many of those it fits. Bacteria and fungi are cells, so a clean slide is a strong statement. The blank being clean rules out the last one.

ASK 16W

Bacteria and fungi are cells. Nothing cell-shaped is on the slide. What does that rule out?

BEHIND THE BUTTON 169W

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TAKES AS READ

some germs are cells that can copy themselves, and viruses cannot copy themselves alone · a test that comes back negative is a result, not a missing one

VERDICT 75W

The negative results do most of the work. Nothing cell-shaped is visible. The marker that finds bacteria is absent. The stain built to find fungal walls is clean, and it rarely misses. The signal only grows where there are living cells to grow in, which fits something that cannot copy itself alone. And the blank is clean, so it did not come out of the room. That is a virus-shaped answer, without naming which one.

TAKEAWAY 14W

What something is gets settled by several tests together, not by one striking picture.

2.2 From sample to a first answer

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 30W

The bench has patient material, healthy tissue and a blank on it. There are four things it could do with them, and two of them use up what they measure.

GUIDE 57W

All four steps will be done, and two of them use up what they measure. Ask of each what has to be true first. Measure something in a dirty sample and you have measured the dirt, close up. And asking whether it can copy itself alone only means something once you know whether it is a cell.

ASK 11W

Put the four steps in the order they have to happen.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

measuring a dirty sample measures the dirt

VERDICT 68W

Each step only means something if the step before it held. The blank and the healthy tissue come first, because they separate the germ from whatever it arrived in. Measure a shape in a dirty sample and you have measured the dirt, close up. Size and shape then say whether this is a cell at all. Only then does asking whether it can copy itself alone mean anything.

TAKEAWAY 14W

A first answer is built out of several observations, not printed by one machine.

2.3 When heat changes the test

MOLECULAR DIAGNOSTICS · A ROOM · CHOICE

SCENE 25W

A test works after being warmed to thirty-seven degrees. After the same part is held at fifty-five, it mostly stops working. Nothing else was changed.

GUIDE 53W

Four options, and only one thing was changed between the two runs. Ask of each whether it names something that was actually varied. The sample was not heated. The thing being looked for was not heated. Nothing else changed at all. So the answer has to be about the part that was warmed.

ASK 6W

What does the temperature comparison show?

BEHIND THE BUTTON 169W

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TAKES AS READ

the working part of a test can be a protein, and a protein has to keep its shape to work

VERDICT 63W

Warming things up usually speeds a reaction up. Everything moves about more and bumps into more. That only holds while the working part keeps its shape. Push the heat far enough and the shape goes, and so does the pocket where the work happens. Here only the heat changed, so the loss belongs to the heat. It says nothing about the patient's sample.

TAKEAWAY 16W

Change one thing and hold the rest, and what changes belongs to the thing you changed.

END OF THE STAGE 21W

Use cell structure, viruses versus cells, prokaryotes and eukaryotes, staining, independent evidence as an evidence chain rather than as isolated facts.

STAGE 3 OF 10

The Point of Entry

BEFORE THE STAGE — FOLLOW

Chen walks a swab from the tent to the lab

A swab passes through four pairs of hands on its way to the laboratory, and every handover is a place germs can get in. Chen, the cell biology lead, walks the whole path this morning. Stay with her, and keep back at the handovers.

PLAN CARD 76W

Day four. The agent reaches the airway in everybody, and only some cells are being damaged. Funke Adeyemi, the cell biologist, has spent two nights on why. Today you work out why those cells and not their neighbours, and design the experiment that separates two steps people keep merging. Your teams start designing a treatment from this picture tonight. If the picture is wrong, they spend three months blocking a step the agent does not need.

BRIEFING 26W

The suspected agent reaches respiratory tissue, but only some cell types become affected. The player must determine how membrane structure, receptors, and transport explain this selectivity.

WHAT YOU DECIDE 18W

Explain why one cell type is vulnerable and select the experiment that separates receptor binding from downstream replication.

3.1 Getting into a cell

CELL & MEMBRANE BIOLOGY · A ROOM · SEQUENCE

SCENE 28W

Only some kinds of cell in the airway are affected. The culture wing has four cards about getting into a cell, and nobody has put them in order.

GUIDE 57W

These four are one chain, so ask what has to happen before each one. Nothing gets in until something on the surface recognises it. A cell without the right piece cannot be got into, however much arrives. That is why only some kinds of cell are affected. Everything after that is downstream of a door already opened.

ASK 9W

Put the four steps of getting in into order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a cell's outer layer lets some things through and keeps others out

VERDICT 60W

The order is physical, and it explains the pattern. Nothing gets in until something on the surface is recognised. A cell without the right piece on its outside cannot be entered, however much lands on it. The layer bends next, because that is what the recognition sets off. Material crosses after that. Everything the cell then does wrongly comes later.

TAKEAWAY 16W

A cell's outside decides what can get in, which is why only some cells are affected.

3.2 Which way things cross

CELL & MEMBRANE BIOLOGY · A PERSON · PROTOCOL

SCENE 26W

Four things from the culture bench, each about something crossing a cell's outer layer. The cell biologist wants each one named before the model goes further.

GUIDE 52W

Four observations and four ways of crossing. Pair them with three questions. Is it going downhill or uphill? Is anything helping it through? And is it too big to fit through anything? The energy being spent is the clue for one of them. The name of the thing crossing tells you nothing.

ASK 9W

Match each observation to how the material is crossing.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

things spread from where there is more of them to where there is less · a cell has to spend energy to move something the other way

VERDICT 63W

Three things tell these apart, and none of them is the name of the molecule. Which way it goes. Downhill is free, uphill costs, and the energy spent is the receipt. Whether it needs help. Something small and greasy slips through, and a charged particle cannot. And how big it is. Something too big for any doorway gets wrapped up and carried in.

TAKEAWAY 17W

How something crosses is worked out from the rules it obeys, not from what it is called.

3.3 Separate landing from getting in

CELL & MEMBRANE BIOLOGY · A PERSON · CONTROL

SCENE 31W

Dr. Maya Chen has a test that counts how much gets into airway cells. Three changes are ready to try, and the treatment meeting needs to know about the first step.

GUIDE 58W

The number you watch is how much gets inside the cells. Three changes are ready. Only one of them acts on the step the meeting is asking about. A change counts only if the number moves more than it wanders between wells. Change one thing, run the test, put it back, and name the change the number follows.

ASK 17W

Which single change makes the entry follow the landing step rather than the dose or the handling?

BEHIND THE BUTTON 149W

What the test measures. The label only shows up when a particle has got inside a cell. Sticking to the outside does not count. That is the point of the test: landing on the cell and getting into it are two different steps, and a treatment that stops one may not stop the other.

Why one of the changes is meant to do nothing. The matched antibody is the same kind of antibody, but it does not target the landing site. Adding it controls for everything else that handling an antibody involves. If the real blocker changes the number and this one does not, the change came from the blocker.

Why the dose is on the list. Add more particles and more get in. The number moves, but nothing has been learned about the step. It is the trap: a change that moves the readout for the wrong reason.

TAKES AS READ

a fair test changes one thing and holds everything else still · an effect that comes back when you undo the change is stronger evidence than a coincidence

VERDICT 67W

Counting the landing places on a cell is only a clue. The real test asks whether the landing step is needed. Block it, keep the dose and the handling the same, and what gets in falls hard. Add a look-alike blocker and nothing changes, so it was not just adding something. Wash the blocker off and it comes back. Both directions is what makes it a cause.

TAKEAWAY 18W

A cause is proved when changing one thing changes the result, and putting it back brings it back.

3.4 What the marquee is taking now

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · SPOT

SCENE 25W

People arrive through the marquee all afternoon. One rule on the whiteboard says who is seen first, and it is rewritten as the ward fills.

GUIDE 33W

Sort arrivals by the rule on the board and leave the rest. The rule changes during the afternoon and nobody calls it out. What counts is the arrivals either side of a change.

ASK 12W

Sort arrivals to the rule on the whiteboard, and keep checking it.

BEHIND THE BUTTON 60W

Why the rule changes. It starts with how ill somebody is. When the isolation beds fill, it becomes who cannot safely wait at home. When the swabs come back, it becomes who is confirmed.

Why the change is where harm happens. Ten minutes on the old rule sends confirmed patients into the queue the new rule was written to clear.

TAKES AS READ

a sorting rule can be changed while people are still arriving

VERDICT 111W

Three rules run across the afternoon and each wants different people. Anybody short of breath, then anybody who cannot stay away from others at home, then anybody with a confirmed test. A person can match two at once, and that is what makes a change cost something real. What the panel scores is the arrivals either side of a change. Most people who come through are wanted by neither rule, so handling those correctly proves nothing. The marquee's own version is the quarter hour after the isolation beds fill. The rule has changed, and the people already being walked down the corridor are being walked there under the rule from before.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

3.5 Signal or noise?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · CALLBACK · CHOICE

SCENE 30W

Seven patients across three hospitals in four days, with the same four symptoms. Dr. Sofia Morales has the printouts, and the press office has asked twice whether to say anything.

GUIDE 64W

Four options, and one of them says nothing can be known at all. Ask of each what it would take to be true. A count on its own is not a signal. It becomes one when you have something to compare it with, and this city normally sees about two a year. Two of the seven were found another way, outside the alert rule.

ASK 17W

The city normally sees about two of these a year. How many were there in four days?

BEHIND THE BUTTON 169W

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TAKES AS READ

a rate counts cases against how many people there are and how long it took · the system only sees the patients who reached a hospital

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A count on its own is not a signal. What makes it one is a comparison with what was expected, written down beforehand. This city logs about two of these a year. Seven in four days is not a busy week. The quiet reading matters most: two of the seven were found by hand, outside the alert that found the other five. If every case had come through one alert, the cluster could have belonged to the alert.

TAKEAWAY 13W

A count becomes a signal only when something says what was expected instead.

END OF THE STAGE 19W

Use phospholipid bilayer, receptors, diffusion, facilitated transport, endocytosis, cell specificity as an evidence chain rather than as isolated facts.

STAGE 4 OF 10

A Test Before Morning

PLAN CARD 67W

Day nine. The hospitals need a working test by sunrise. Patel's overnight run flagged several patients positive — and one of the controls, a tube that had nothing in it. Three thousand people have already been told to stay home on that run. The bulletin goes out at six. Today you decide whether the run can be believed, and what a positive would mean for one patient.

BRIEFING 26W

Hospitals need a rapid diagnostic by sunrise. The molecular signal is promising, but contamination, false negatives, and prevalence could make a technically impressive test clinically misleading.

WHAT YOU DECIDE 13W

Approve a diagnostic workflow only after its analytical and clinical limitations are explicit.

4.1 A run you can trust by morning

MOLECULAR DIAGNOSTICS · A ROOM · SEQUENCE

SCENE 37W

Hospitals want a test by sunrise and the room has four hours. Four steps are on the whiteboard. The machine multiplies whatever is in the tube a billion times, which is why the order is argued about.

GUIDE 55W

All four steps will happen, so ask what each one protects. Whatever is in the tube gets multiplied, including anything that drifted in from the room. Whose sample it is has to survive the first step. And the controls have to be in the run, not added afterwards. That is why this is about order.

ASK 9W

Put the four steps of the run in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

the machine copies whatever is in the tube, wanted or not

VERDICT 65W

A patient result read before the controls is a machine reading, not evidence. The name has to survive the first step, or the answer belongs to nobody. The controls go in with the samples. Added later, they test a different run. One shows the chemistry worked. Another shows the signal did not come out of the room. Only then is the patient's tube worth reading.

TAKEAWAY 14W

A test result is what a controlled process produced, not what a machine displayed.

4.2 Why did the test turn positive?

MOLECULAR DIAGNOSTICS · A PERSON · TRACE

SCENE 32W

Dr. Ravi Patel has patient tubes, two controls and a separate machine on one screen. One ward has already cleared isolation rooms. If this run is wrong, patients get moved before morning.

GUIDE 47W

Open each result to see what its run depended on. Keep the ones that would still stand if one step of the process were contaminated, and untick the rest. Then name the step the failing ones share. A ward is already clearing rooms, so both halves matter.

ASK 10W

Which results share one path, and what evidence survives it?

BEHIND THE BUTTON 133W

How one shared step makes results agree wrongly. Tubes handled in the same batch go through the same liquids, the same machine and the same hands. Contamination anywhere in that chain shows up in every tube after it, and it looks exactly like a cluster of real positives.

What the controls do. A negative control that stays negative says the whole batch was not contaminated. A positive control says the chemistry worked. Neither can catch contamination that arrived after the controls were set, which is why the separate machine is on the screen.

Why a repeat is not enough. Running the same test again on the same tube goes through the same steps, so a contaminated result repeats perfectly. A different machine on a different sample is the evidence that survives the question.

TAKES AS READ

a blank tube has no patient sample in it · two machines with different chemistry fail in different ways

VERDICT 71W

A blank tube has no patient material in it. So a positive there came from the room or the chemicals. The patient tubes went through the same room and the same chemicals. That makes their positives doubtful, however many there are. The control that was meant to be positive shows the chemistry can work, and it cannot rescue a spoiled run. The other machine used different chemistry, so its result stands.

TAKEAWAY 12W

Agreement is not evidence when everything agreeing came down the same pipe.

4.3 What does a positive mean?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · BALLPARK

SCENE 33W

The mayor's office wants to test the whole town, and to treat every positive as a case. The desk has a table from ten thousand people, with the true answer beside the test's.

GUIDE 50W

Five numbers, and two of them are the whole town split into ill and well. Those were used to work out the flags. Ask which two this share needs. A share compares part with a whole. The whole here is everybody the test flagged, not everybody who is really ill.

ASK 11W

Out of every hundred positive results, how many are real cases?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

how good a test is does not change with how common the illness is

VERDICT 77W

The test is good. It catches ninety of the hundred people who really have it, and it correctly clears ninety-nine per cent of everyone else. But everyone else is nine thousand nine hundred people, and one per cent of that is ninety-nine wrong positives. So there are ninety right positives and ninety-nine wrong ones. Less than half the positives are real. Nothing is wrong with the test. What makes it happen is that the illness is rare.

TAKEAWAY 15W

When something is rare, most positives can be wrong even with a very good test.

4.4 What kind of germ fits everything?

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · CALLBACK · CHOICE

SCENE 33W

Pathology has four results from the same samples and still cannot name what this is. The samples are waiting under the hood. Ortiz needs a broad answer before the overnight run is booked.

GUIDE 53W

Four options, and three results on the bench are negatives. Nothing cell-shaped on the slide. No bacterial marker. No fungal wall. Ask of each option how many of those it fits. Bacteria and fungi are cells, so a clean slide is a strong statement. The blank being clean rules out the last one.

ASK 16W

Bacteria and fungi are cells. Nothing cell-shaped is on the slide. What does that rule out?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

some germs are cells that can copy themselves, and viruses cannot copy themselves alone · a test that comes back negative is a result, not a missing one

VERDICT 75W

The negative results do most of the work. Nothing cell-shaped is visible. The marker that finds bacteria is absent. The stain built to find fungal walls is clean, and it rarely misses. The signal only grows where there are living cells to grow in, which fits something that cannot copy itself alone. And the blank is clean, so it did not come out of the room. That is a virus-shaped answer, without naming which one.

TAKEAWAY 14W

What something is gets settled by several tests together, not by one striking picture.

END OF THE STAGE 18W

Use PCR logic, controls, sensitivity, specificity, predictive value, prevalence as an evidence chain rather than as isolated facts.

Why Are Some Patients Sicker?

BEFORE THE STAGE — HUNT

Six ventilators, and the board finds four

Before tonight the ward has to know what it has. Four are where the board says. Two were moved at the weekend and nobody wrote it down. A machine nobody can find does not exist at two in the morning.

PLAN CARD 70W

Day thirteen. Nakamura has two patients who arrived with the same complaint and the same results. One is walking to the bathroom. The other is failing, and their oxygen reading looks normal. Today you find which link in the oxygen chain has broken. Nakamura has one line and two patients she would like it in. Treat the wrong link and the treatment is aimed at a part that is working.

BRIEFING 25W

Patients with similar test results have radically different oxygen levels and recovery times. The team must connect cellular damage to organ-system physiology and homeostatic failure.

WHAT YOU DECIDE 16W

Identify the physiological bottleneck and select measurements that distinguish ventilation, diffusion, circulation, and cellular oxygen use.

5.1 Where the oxygen stops getting through

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · CHAIN

SCENE 31W

A patient's oxygen is dropping while Dr. Lena Ortiz reads the bedside screen. The blood carries plenty. The heart is pumping normally. The new lung pictures are not normal at all.

GUIDE 55W

Build the path oxygen takes, in order, from the air in the room to the tissues that use it. Then name the link where it is failing for this patient. Read the screen rather than guessing: the blood is carrying plenty and the heart is pumping normally, and each of those rules a link out.

ASK 8W

Where is the oxygen path most likely failing?

BEHIND THE BUTTON 127W

What the path is. Air comes into the lungs. Oxygen crosses from the tiny air sacs into the blood. The blood carries it, the heart moves it around, and the tissues take it up. Each step can fail on its own, and each failure needs different treatment.

How the numbers rule links out. Plenty of oxygen carried in the blood means the carrying step is fine. A heart pumping normally means the delivery step is fine. So the failure has to be somewhere those two do not cover.

Why the lung pictures matter most. They point at the step where oxygen crosses from the air into the blood. That is the one link the other measurements cannot clear, and it is treated very differently from the others.

TAKES AS READ

oxygen has to go from the air into the lungs, into the blood, and out to the body, in that order · a normal reading at one step makes a simple failure there less likely

VERDICT 71W

Oxygen delivery is a chain, so one normal reading proves nothing on its own. The breathing out is near normal, so the patient is not simply under-breathing. The blood carries a normal amount, so there is no shortage of carriers. The heart is moving a normal amount of blood, so the pump is not the problem. What is left is the handover from air to blood, and the lung pictures agree.

TAKEAWAY 21W

A chain is found by checking every handover, and the normal ones narrow it down as much as the odd ones.

5.2 Carrying is not delivering

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · BALLPARK

SCENE 32W

The bedside argument has stalled on which of two patients is worse. Each litre of their blood carries the same oxygen. One moves three litres a minute and the other moves five.

GUIDE 52W

Five numbers, and three belong elsewhere. The other patient's pump, how full the blood is, and how much carrier there is. Ask of each whether this sum needs it. Both patients carry the same oxygen in every litre, so that cannot tell them apart. What differs is how fast the blood moves.

ASK 12W

How much oxygen a minute reaches the patient with the slower pump?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

how much arrives depends on how much each litre carries and how many litres arrive

VERDICT 75W

Two patients whose blood carries exactly the same oxygen can differ by a factor of two in what reaches them, because the flow is doing the work. Three litres a minute at a fifth of a litre of oxygen each is 0.6 a minute. Five litres is a full litre. The bedside readings everybody quotes describe what each litre carries. Nobody can read delivery off them, because delivery is a rate and those are not.

TAKEAWAY 14W

A full tank means nothing until something says how fast it is being delivered.

5.3 Which measurement is still missing

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A PERSON · CHOICE

SCENE 31W

The oxygen path points at the lungs, and the team wants one more measurement before rounds. Three of the four on offer have already been taken, in one form or another.

GUIDE 52W

All four options are real measurements. Ask of each whether the team already has it, in one form or another. Three of these repeat something on the chart. Repeating the same reading on three machines tells you about the machines. One option looks somewhere nobody has looked yet, further along the path.

ASK 7W

Which measurement would actually add something here?

BEHIND THE BUTTON 169W

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What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a measurement adds nothing if the chart already answers it

VERDICT 83W

Three of these have already been answered. The oxygen and carbon dioxide are on the chart from the last panel. The pump has been measured and read normal. Three machines agreeing about the same number tells you the machines agree, which nobody doubted. What has not been measured is what is happening downstream, in the body's own tissues. That is where a delivery problem shows up as harm, and it is the only reading here that could still change what the team does.

TAKEAWAY 14W

The measurement worth taking is the one nobody can already answer from the chart.

5.4 Clean side, dirty side

ECOLOGY & ONE HEALTH · ONE HEALTH FIELD STATION · A ROOM · BELT

SCENE 37W

The courier leaves in ten minutes and the tray has to be sorted. One side of the bench has touched nothing outside the cabinet. The other has been open to the room, a glove, or a patient.

GUIDE 33W

Two bins. Ask where the thing has been since it was last clean. If it has been open to the room, it cannot go back to the clean side, however clean it looks.

ASK 11W

Send each item to the side its history puts it on.

BEHIND THE BUTTON 59W

Why looking clean is not the test. Germs are too small to see. Once something has been open to the room it can carry what the room carries, and it looks exactly the same.

Why it is worth ten minutes. One dirty item put back on the clean side puts the room's germs into every sample that follows it.

TAKES AS READ

something can be clean and stop being clean without looking different

VERDICT 110W

The clean side is about where a thing has been. Not about how it looks. Something kept inside the running cabinet has only met filtered air. So has something still sealed. Anything that has met the room can carry what the room carries. And it looks exactly the same afterwards. That is why a sealed swab dropped on the floor is dirty on the outside. It is also why an open plate that sat in a working cabinet for an hour is still clean. Getting it wrong does not spoil one sample. It puts the room's germs into every sample that follows. Later, nobody can tell which results were real.

TAKEAWAY 14W

What decides clean is where a thing has been, not what it looks like.

5.5 When heat changes the test

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · CHOICE

SCENE 25W

A test works after being warmed to thirty-seven degrees. After the same part is held at fifty-five, it mostly stops working. Nothing else was changed.

GUIDE 53W

Four options, and only one thing was changed between the two runs. Ask of each whether it names something that was actually varied. The sample was not heated. The thing being looked for was not heated. Nothing else changed at all. So the answer has to be about the part that was warmed.

ASK 6W

What does the temperature comparison show?

BEHIND THE BUTTON 169W

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TAKES AS READ

the working part of a test can be a protein, and a protein has to keep its shape to work

VERDICT 63W

Warming things up usually speeds a reaction up. Everything moves about more and bumps into more. That only holds while the working part keeps its shape. Push the heat far enough and the shape goes, and so does the pocket where the work happens. Here only the heat changed, so the loss belongs to the heat. It says nothing about the patient's sample.

TAKEAWAY 16W

Change one thing and hold the rest, and what changes belongs to the thing you changed.

END OF THE STAGE 19W

Use homeostasis, gas exchange, circulation, oxygen transport, feedback, organ-system integration as an evidence chain rather than as isolated facts.

STAGE 6 OF 10

The Hidden Transmission Route

PLAN CARD 73W

Day twenty-four, and Morales gets her turn. Ward isolation has held for nine days, and cases keep appearing in streets with nobody in hospital. Grace Oduya kept a sewer sample from the north of the city by chance. It was positive four days before the first patient walked in. Today you decide where this is really being passed on, and where to watch. Oduya has three teams and the city has four drains.

BRIEFING 17W

Cases continue despite hospital isolation. Wastewater and neighborhood data suggest transmission is occurring before severe symptoms appear.

WHAT YOU DECIDE 17W

Identify the hidden route and target surveillance where it changes decisions rather than where sampling is easiest.

6.1 What each stream is looking at

IMMUNOLOGY & TREATMENT · A ROOM · PROTOCOL

SCENE 32W

Isolation has held for nine days and cases keep appearing. Four streams of information are on the table at the water plant, and each one saw a different slice of the city.

GUIDE 60W

Four streams, and each one looked at a different slice of the city. Pair them by asking who is in that group, and what the stream can and cannot show. A reading from the sewer does not name a person, and it does not say how the germ travels. And a ward stays full while people are still getting better.

ASK 8W

Match each stream to what it actually supports.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

who ends up in a set of numbers is decided by how they were collected

VERDICT 80W

Each of these watches a different set of people. The sewage watches everybody, whether or not they went to a hospital, which is why it can move first. It cannot name a person, and it does not prove the water is how anybody caught it. Interviews ask people directly and depend on what they remember. A volunteer survey samples volunteers. And the number of people ill now stays high after new cases fall, because people stay ill for a while.

TAKEAWAY 17W

Each stream watches a different group of people, and what it can show follows from which group.

6.2 Same instructions, different cells

CELL & MEMBRANE BIOLOGY · A PERSON · CHOICE

SCENE 29W

The stain shows the landing site all over the airway cells, and nowhere on the muscle cells two millimetres away. Same patient, same slice, same instructions inside every cell.

GUIDE 56W

Four options, and both kinds of cell came from one patient, one slice, stained together. Ask of each whether having an instruction and using it are the same thing. Every cell holds all of them. So what has to differ is which ones get read, and that is what makes an airway cell an airway cell.

ASK 16W

Every cell has the same instructions. Why do only some of them make the landing site?

BEHIND THE BUTTON 169W

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Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

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TAKES AS READ

every cell in a body carries the same set of instructions

VERDICT 81W

Every cell carries the whole set of instructions, so having the instruction explains nothing about where the thing gets made. What decides it is which instructions a cell is reading. That is what makes an airway cell an airway cell rather than a muscle cell. For an outbreak it is the practical point. The germ can only get into cells that make its landing site, which is why this one causes an illness of the airways rather than of the muscles.

TAKEAWAY 13W

Every cell carries the same instructions and reads a different part of them.

6.3 What counts as a case

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · CALLBACK · SEQUENCE

SCENE 36W

Three hospitals start counting at six in the morning. Each has written its own idea of what counts. Dr. Lena Ortiz, who leads the ward side, wants one on every desk before the night shift leaves.

GUIDE 61W

All four steps will happen. Ask of each what it lets you check. A rule for what counts is a tool, and it can be wrong in two directions. Too narrow and you miss the spread. Too wide and it fills up with ordinary coughs. So it gets tried on known cases and on clear non-cases, and those catch opposite mistakes.

ASK 13W

Put the four steps in the order each one can be checked in.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a graph of cases only means something if everybody counted the same thing · a definition can be too narrow or too wide, and both are wrong

VERDICT 73W

A definition of what counts is an instrument, and it can be wrong in two directions. Too narrow and most of the spread is invisible. Too wide and the count fills up with ordinary autumn coughs. That is why it gets tested twice before it goes out. Against patients everybody agrees are cases, which catches one that is too narrow. Against people who obviously are not, which catches one that is too wide.

TAKEAWAY 14W

A definition turns a worry into something two people can count the same way.

END OF THE STAGE 19W

Use incidence, prevalence, transmission chains, sampling, R concepts, asymptomatic spread as an evidence chain rather than as isolated facts.

STAGE 7 OF 10

The Animal Connection

BEFORE THE STAGE — CANVASS

Who in Ferry Ward is staying home

The spread estimate assumes people followed the advice. A ward that stayed home and a ward that could not look the same on paper. Ask along the streets until enough answers have come back to tell them apart.

PLAN CARD 69W

Day thirty-three. Readings from three patients look close to something found in animals along the river. The council has signed an order to clear the heron colony on Saturday. Mateo Castellanos has been out there twice and says the herons are the wrong animal. Today you decide what part any animal is really playing, and design the sampling that could show it. Nearby is not the same as guilty.

BRIEFING 28W

Genetic similarities and field reports point toward an animal reservoir near the river corridor. The player must investigate without assuming that the most visible species is the source.

WHAT YOU DECIDE 12W

Build a defensible reservoir hypothesis using ecological exposure, prevalence, and genetic relatedness.

7.1 Carrier, carrier-of-it, or bystander

ECOLOGY & ONE HEALTH · ONE HEALTH FIELD STATION · A ROOM · PROTOCOL

SCENE 31W

Sequences from three patients sit close to something found in animals along the river. The field station has four patterns on the board, and a trapper waiting outside with more animals.

GUIDE 53W

Four patterns and four roles. Pair them by asking what the animal does, not how close it is. Does the germ live in it? Does it pass it on? Does the chain stop there? One animal is simply common near cases and always tests clean. Sharing the ground is not sharing the germ.

ASK 10W

Match each pattern to the role it makes an animal.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

an animal can carry a germ, move it about, or simply be nearby

VERDICT 95W

A role is a claim about what keeps happening, not about which animal people are afraid of. An animal that carries the germ for a long time without getting very ill is where it lives between outbreaks. Something that bites and moves it from one animal to another is a carrier of it, whether or not it is where the germ lives. An animal that gets ill and passes it to nobody is where the chain stops. And an animal that is simply always nearby and always tests clean shares the ground, not the germ.

TAKEAWAY 17W

What an animal's role is comes from what it does over time, not from being close by.

7.2 Test where it lives

ECOLOGY & ONE HEALTH · ONE HEALTH FIELD STATION · A PERSON · SEQUENCE

SCENE 29W

Trappers report sick animals on the flood plain. Nobody has sampled in a way that could tell where the germ lives from an animal that caught it from people.

GUIDE 58W

All four steps will happen, and the campaign gets paid for once. A positive animal is not where the germ lives. What makes it that is a search that could have found it somewhere else and did not. Ask of each step what makes the next one mean anything. Comparison places are what make a figure worth having.

ASK 10W

Put the four steps of the field campaign in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

where you look decides what you can find

VERDICT 62W

One positive animal proves nothing about where a germ lives. What makes it evidence is a design that could have found it elsewhere and did not. So the order matters. Where people got ill says which animals to sample. Comparison places make the numbers mean something. Only then does comparing animals with people say anything, and only then can it be tested.

TAKEAWAY 16W

A claim about where something lives needs a design that could have found it somewhere else.

7.3 Choose the field campaign

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · ALLOCATE

SCENE 25W

Dr. Arjun Singh has one field campaign and twenty sampling slots. Likely animals, comparison places and a second visit later all want the same crews.

GUIDE 51W

One campaign, twenty sampling slots, and more animals, places and visits than the crews can cover. Watch the list of questions rather than the map. It shows which comparisons your plan can still make, and a plan that samples the most animals while keeping no comparison cannot tell the stories apart.

ASK 11W

Which plan keeps the comparisons that could tell the stories apart?

BEHIND THE BUTTON 128W

Why comparisons rather than counts. Finding the germ in an animal shows it can carry it. It does not show that animal is the source. That needs a contrast: this animal against others, this place against another place, now against later.

What a comparison place is for. If the germ turns up everywhere, the finding says nothing about this outbreak. Finding it here and not there is what connects the animals to the people who got ill, and it costs half the campaign to be able to say so.

Why a second visit is worth so much. A single visit is a snapshot. It cannot tell a settled source from a one-off, and that difference decides whether anything has to change about how people live near these animals.

TAKES AS READ

a sample stands for the group it was drawn from · telling where a germ lives from where it merely arrived needs a comparison

VERDICT 58W

A positive animal near a sick person fits several stories. The germ lives in that animal. It caught it from people. Or both met the same thing. Only comparisons can separate those. The same animal far from any cases tests the place. Other animals in the same habitat test the species. Coming back later tests whether it stays.

TAKEAWAY 17W

A field plan is defined by the explanations it can rule out with the samples it chooses.

7.4 Getting into a cell

CELL & MEMBRANE BIOLOGY · A ROOM · CALLBACK · SEQUENCE

SCENE 28W

Only some kinds of cell in the airway are affected. The culture wing has four cards about getting into a cell, and nobody has put them in order.

GUIDE 57W

These four are one chain, so ask what has to happen before each one. Nothing gets in until something on the surface recognises it. A cell without the right piece cannot be got into, however much arrives. That is why only some kinds of cell are affected. Everything after that is downstream of a door already opened.

ASK 9W

Put the four steps of getting in into order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a cell's outer layer lets some things through and keeps others out

VERDICT 60W

The order is physical, and it explains the pattern. Nothing gets in until something on the surface is recognised. A cell without the right piece on its outside cannot be entered, however much lands on it. The layer bends next, because that is what the recognition sets off. Material crosses after that. Everything the cell then does wrongly comes later.

TAKEAWAY 16W

A cell's outside decides what can get in, which is why only some cells are affected.

END OF THE STAGE 20W

Use reservoirs, vectors, food webs, host range, ecological sampling, One Health as an evidence chain rather than as isolated facts.

STAGE 8 OF 10

Treatment Failure

BEFORE THE STAGE — EVADE

The reporter at the gate who wants today's number

He has been there since six, wanting the case count before it is checked. Nothing goes out before the count is reconciled. Keep clear and finish your round, because a wrong number takes a week to correct.

PLAN CARD 78W

Day sixty. The treatment that bought this city six weeks is failing in one patient in four. The failures cluster on the wards using it most. Ortiz wants the dose raised today. Lena Bergström has a resistant sample banked a month before the first dose was given anywhere. Today you work out why it is failing, and choose something that does not make it worse. Raising the dose is the fastest move available, and may be the worst.

BRIEFING 30W

A therapy that initially helped is failing in a growing subset of patients. The player must determine whether the cause is biological resistance, poor delivery, incorrect dosing, or patient heterogeneity.

WHAT YOU DECIDE 18W

Separate inherited resistance from non-genetic treatment failure and propose a strategy that reduces selective advantage without undertreating patients.

8.1 Why did the treatment stop working?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · DIAGNOSIS

SCENE 31W

A treatment that worked is failing, and another patient got worse overnight. The ward has the drug levels, the counts over time, and the germ's own readout from before and after.

GUIDE 59W

Five readings, and the ordinary ones do the work. The drug got to a level that works. The count fell. The doses were all given. Ask of each option how many of the five it fits. What happened next is a rebound while the patient was still taking it. The reference sample staying sensitive says the test is fine.

ASK 6W

Which explanation fits the whole panel?

BEHIND THE BUTTON 156W

Why the quiet readings matter. It is easy to build a story around the number that is out of range. But most wrong explanations fail on something that did not move. If a cause would have pushed a second reading up and that reading is where it should be, the cause is not what happened, however well it explains the first one.

How to use the candidates. Take each one and ask what the panel would look like if it were true. Then compare that with the panel in front of you. Doing it in that order stops you fitting the story to the numbers, which is easy to do without noticing.

Why there is only one answer. More than one candidate will fit part of the panel, and that is on purpose. Only one fits every part of it. If two still look possible, find the reading they disagree about and let that reading choose.

TAKES AS READ

a drug only works if it reaches the germ in a large enough amount

VERDICT 80W

Treatment can fail in several ways, and the normal readings say which. The drug reached the right level, and the count fell, so the drug was working and reaching its target. Then the count came back up while the patient was still taking it, and the changed form of the germ took over during the same weeks. The reference sample, run alongside, is still knocked down by the drug. So the drug still works. This germ no longer minds it.

TAKEAWAY 16W

Something living can change while you watch, if what you are doing decides which ones survive.

8.2 How resistance spreads

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 35W

The treatment worked for six weeks, and now fails in one patient in four. The failures cluster on the wards where it has been used longest. Somebody has said the germ learned to beat it.

GUIDE 57W

These four are one chain, and somebody has said the germ learned. Ask of each card what has to be true before it. A few of them are already different before any drug arrives. What the drug changes is which ones are left to multiply. The wards where it has been used longest have simply had longer.

ASK 6W

Put the four steps in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a population is never all the same, before anything acts on it

VERDICT 78W

A treatment does not teach a germ anything, and it does not make the form it needs. That form was already there, rare, before the drug arrived. What the drug changes is which ones have offspring. It kills the ones it can, the few it cannot survive, those few multiply, and after enough rounds most of what is left is the form the drug cannot touch. The clustering on the long-treated wards is that process having run longer.

TAKEAWAY 9W

Nothing learns. What changes is which ones leave descendants.

8.3 One thing before the prescriptions change

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · CHOICE

SCENE 29W

The treatment worked in eighty-two patients in a hundred nine weeks ago, and in fifty-one now. Eleven of the failures had their doses recorded and their drug levels checked.

GUIDE 59W

Four options, and a treatment can fail in three places. Not given properly. Not taken up by the body. Or no longer working where it acts. Ask of each which of the three it tests. Recorded doses and checked levels already weaken the first two. And matched patients who got better are what make the last option a test.

ASK 11W

What is the one thing to do before anybody's prescription changes?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a comparison group is what turns a finding into evidence

VERDICT 83W

A treatment can fail in three places. It can be given wrongly, it can fail to arrive, or it can arrive and no longer work. Recorded doses and measured levels rule out the first two here. So the change is probably in the germ. Reading the germ from the failures would show a change, and a change on its own proves nothing. Reading it from patients who got better as well is what shows the change tracks the failures rather than being everywhere.

TAKEAWAY 16W

Compare the failures with matched successes, or the finding is just a description of the failures.

8.4 Keeping the air going the right way

IMMUNOLOGY & TREATMENT · A ROOM · HOLD

SCENE 33W

The isolation room has to stay at lower pressure than the corridor, so air moves in rather than out. Staff come and go, and a door is propped while a patient is turned.

GUIDE 42W

Hold the pressure difference inside the band on the gauge. The band narrows as the ward fills. Every door that opens keeps letting pressure away for as long as it is open, so set the fan to answer it rather than waiting.

ASK 10W

Hold the room below the corridor while the shift runs.

BEHIND THE BUTTON 69W

Why lower pressure. Air moves from higher pressure to lower. A room kept below the corridor pulls air in under the door instead of pushing it out, so what is in the room stays there and leaves through a filter.

Why a propped door keeps costing. It is not one drop and done. It leaves a path open, and the pressure keeps leaking away until the fan works harder.

TAKES AS READ

air moves from higher pressure towards lower pressure

VERDICT 145W

Air moves from higher pressure to lower. A room held below the corridor draws air in under its door, so what is in the room stays in the room and leaves through the filter. Everything in this shift works against that, and none of it is a single push. A propped door leaves a path open and the pressure keeps leaking away while it stands. So the fan is set to a new speed and left there rather than nudged. The band narrows as the ward fills, because a difference that is just about enough with two patients in the room is not enough with six. And the failure is quiet. A room that has gone the wrong way is pushing its air into the corridor and looks exactly like a room that has not, which is why somebody reads the gauge instead of assuming.

TAKEAWAY 10W

A leak you do not answer does not close itself.

8.5 Which way things cross

CELL & MEMBRANE BIOLOGY · A ROOM · CALLBACK · PROTOCOL

SCENE 26W

Four things from the culture bench, each about something crossing a cell's outer layer. The cell biologist wants each one named before the model goes further.

GUIDE 52W

Four observations and four ways of crossing. Pair them with three questions. Is it going downhill or uphill? Is anything helping it through? And is it too big to fit through anything? The energy being spent is the clue for one of them. The name of the thing crossing tells you nothing.

ASK 9W

Match each observation to how the material is crossing.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

things spread from where there is more of them to where there is less · a cell has to spend energy to move something the other way

VERDICT 63W

Three things tell these apart, and none of them is the name of the molecule. Which way it goes. Downhill is free, uphill costs, and the energy spent is the receipt. Whether it needs help. Something small and greasy slips through, and a charged particle cannot. And how big it is. Something too big for any doorway gets wrapped up and carried in.

TAKEAWAY 17W

How something crosses is worked out from the rules it obeys, not from what it is called.

END OF THE STAGE 17W

Use resistance, selection pressure, dose-response, targets, pharmacokinetics, heterogeneity as an evidence chain rather than as isolated facts.

Design the Intervention

PLAN CARD 76W

Day seventy-eight. The city wants something done, and the room is arguing as though it can only have one thing. The favourite treatment sticks to the exact patch that changed in the north, which Bergström has now said three times. Today you choose what to fund, and say what evidence is needed before any of it reaches a person. One idea is ready on Monday. One takes a year. That is the argument for having several.

BRIEFING 30W

The city needs an intervention that can be manufactured, tested, and deployed. The player must compare vaccines, targeted therapies, and nonpharmaceutical measures using evidence about mechanism and evidence about populations.

WHAT YOU DECIDE 12W

Select a layered intervention portfolio and identify the evidence required before deployment.

9.1 Where each measure acts

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · PROTOCOL

SCENE 31W

The city wants something done, and the room is arguing as if only one thing can be chosen. Four proposals are on the table, and each acts at a different moment.

GUIDE 55W

Four aims and four measures. Pair them by asking when each one acts. Before anybody meets it? As it gets in? After somebody is already ill? Or between people? The room is arguing as if only one can be chosen. Each of them fails in a different way, which is the argument for having several.

ASK 9W

Match each aim to the measure that achieves it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

an illness has a before, a during and an after, and different tools act at each

VERDICT 83W

Each of these acts at a different moment, which is why they add up instead of competing. Stopping the germ getting into a cell has to happen before it does. Calming the damage happens afterwards, and giving it too early gets in the way of a defence still doing its job. Building up defences in advance is what a vaccine does, and arriving first is its whole value. Ventilation and distance act on the passing on, whatever the germ turns out to be.

TAKEAWAY 16W

Measures that act at different moments are stronger together than more of any one of them.

9.2 From a promising result to a fair trial

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 38W

Two things look promising in the dish, and the city wants one of them in patients within the month. Neither has been tried in an animal, and the plan on the table says nothing about when to stop.

GUIDE 54W

All four steps will happen, and hurrying changes their speed rather than their order. Ask of each what the next step would mean without it. Saying what it should do comes before finding out if it does. And without a rule for when to stop, a trial cannot be stopped when it should be.

ASK 6W

Put the four steps in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

working in a dish is not evidence of helping a person

VERDICT 79W

Being in a hurry changes the speed of this and not its shape. Knowing what the thing is supposed to do comes first, because otherwise there is nothing to test it against. Animals and laboratory work come next, because a thing that harms an animal has told you something before a person takes it. Then comes what would count as working, and when to stop. Decide those after the results arrive and the trial can prove anything you like.

TAKEAWAY 15W

Being in a hurry changes how fast the steps go, not which steps there are.

9.3 Three slots, six good ideas

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · ALLOCATE

SCENE 31W

Dr. Nia Okafor has six workable ideas and three slots to fund. Some can work this month. Others take longer, and some fail for exactly the same reasons as each other.

GUIDE 51W

Six workable ideas, three slots. Watch which response questions your current three can still answer. Ideas that fail for the same reason look like two defences and behave like one, so a set that holds when one assumption turns out wrong is worth more than three strong ideas that fall together.

ASK 12W

Which three keep the response working if one assumption turns out wrong?

BEHIND THE BUTTON 116W

What layers are for. No single measure works perfectly, so you use several whose weaknesses are different. Two ideas that both depend on people reporting symptoms quickly are not two layers. They are one, paid for twice.

Why timing splits them too. Some can work this month and some take a season to set up. A set made only of slow ones leaves the next four weeks uncovered, whatever it achieves later.

How to test a set. Take each assumption in turn — people follow the advice, supplies arrive, the germ behaves as expected — and ask what still works if it is wrong. The set that survives each single failure is the one to fund.

TAKES AS READ

things that work in different ways fail for different reasons · a plan with three slots gives up three good ideas

VERDICT 57W

Layers are only layers if they fail separately. Ventilation lowers the chance of catching it, without waiting for anybody to be found. Fast isolation works after somebody is found, and cuts what happens next. A vaccine takes longer and protects in a different way. Treatment and sick pay both matter, and three slots cannot buy six things.

TAKEAWAY 18W

A set of measures is stronger when they do not all depend on the same thing going right.

9.4 Separate landing from getting in

CELL & MEMBRANE BIOLOGY · A ROOM · CALLBACK · CONTROL

SCENE 31W

Dr. Maya Chen has a test that counts how much gets into airway cells. Three changes are ready to try, and the treatment meeting needs to know about the first step.

GUIDE 58W

The number you watch is how much gets inside the cells. Three changes are ready. Only one of them acts on the step the meeting is asking about. A change counts only if the number moves more than it wanders between wells. Change one thing, run the test, put it back, and name the change the number follows.

ASK 17W

Which single change makes the entry follow the landing step rather than the dose or the handling?

BEHIND THE BUTTON 149W

What the test measures. The label only shows up when a particle has got inside a cell. Sticking to the outside does not count. That is the point of the test: landing on the cell and getting into it are two different steps, and a treatment that stops one may not stop the other.

Why one of the changes is meant to do nothing. The matched antibody is the same kind of antibody, but it does not target the landing site. Adding it controls for everything else that handling an antibody involves. If the real blocker changes the number and this one does not, the change came from the blocker.

Why the dose is on the list. Add more particles and more get in. The number moves, but nothing has been learned about the step. It is the trap: a change that moves the readout for the wrong reason.

TAKES AS READ

a fair test changes one thing and holds everything else still · an effect that comes back when you undo the change is stronger evidence than a coincidence

VERDICT 67W

Counting the landing places on a cell is only a clue. The real test asks whether the landing step is needed. Block it, keep the dose and the handling the same, and what gets in falls hard. Add a look-alike blocker and nothing changes, so it was not just adding something. Wash the blocker off and it comes back. Both directions is what makes it a cause.

TAKEAWAY 18W

A cause is proved when changing one thing changes the result, and putting it back brings it back.

END OF THE STAGE 20W

Use antigens, immune memory, drug targets, intervention layers, efficacy versus effectiveness as an evidence chain rather than as isolated facts.

STAGE 10 OF 10

The Final Briefing

BEFORE THE STAGE — TAG

Catch Patel before he goes into the container lab

Patel, the molecular diagnostics lead, has the sheet that says which samples were tested twice. Once he is gowned up he is gone for three hours, and this morning's positives cannot be read without knowing which are repeats.

PLAN CARD 74W

Day one hundred and forty. The mayor is on air in an hour, and behind that broadcast sit five months of conclusions that are not equally solid. The route is settled. The animal source is not, and Castellanos has said so in writing. Today you go through the claims one at a time, and decide what the city keeps funding after the emergency money closes. Say too little and the next outbreak arrives unwatched.

BRIEFING 21W

The mayor, hospitals, and public must receive one integrated account of what happened, what remains uncertain, and which actions should continue.

WHAT YOU DECIDE 13W

Produce a claim-by-claim evidence package with explicit residual uncertainties and durable surveillance responsibilities.

10.1 What each claim has earned

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · ATTEST

SCENE 27W

The emergency is closing, and four signed claims are going into the city's record. Two checks are left. The evidence under the four is not equally strong.

GUIDE 62W

Four signed claims are going into the city's record and two checks are left. Open each claim and read what backs it now. Some rest on measurements from this outbreak, some on a chain of reasoning, and some on how confidently they are written. Hold what the evidence does not support, and check where a wrong claim would do the most damage.

ASK 13W

Which claims need checking before the report can say them at full strength?

BEHIND THE BUTTON 126W

Why the record raises the bar. People will use these claims later without being able to see the evidence behind them. A claim taken back afterwards has already done its work, and the correction rarely travels as far as the claim did.

What full strength means. The same finding can be written as 'this is what happened', or 'this fits what happened', or 'we cannot rule it out'. Checking decides which of those it has earned, so a claim can pass by being written more carefully instead.

Why choosing is the real skill. Two checks and four claims means picking. Ask which one, if it is wrong, would send the next response in the wrong direction. That is not always the one with the weakest evidence.

TAKES AS READ

a signature records that somebody said it, not that anybody checked it · a strong claim can survive one of its sources being wrong

VERDICT 66W

A report should not give everything the same confidence. How it spreads is backed by three separate streams, so it can carry planning. Who gets sickest comes from one small group of patients, and it is about to become city policy. The treatment claim covers eight weeks and is written as though it covers longer. The animal question is open, and can be written as open.

TAKEAWAY 11W

Say each claim at the strength its own evidence can carry.

10.2 Build the public explanation

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 31W

The mayor goes live in an hour. How it spreads is settled. The animal question is not. The treatment has eight weeks behind it. Every sentence will be replayed for years.

GUIDE 63W

All four things will be said, so the order is what is being chosen. Ask of each how much its evidence can carry. Three streams that agree? One small group of patients? Or a question still open? Every sentence gets replayed for years. Saying the open part as firmly as the settled part is what makes the whole briefing hard to believe later.

ASK 10W

Order the briefing by how much each claim can carry.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

people judge a claim by how it was supported, once they find out

VERDICT 64W

Order it by what the evidence carries, strongest first. That is the order in which a listener can tell one kind of claim from another. Say the animal question as firmly as the spreading and the whole thing falls apart the day the animal turns out to be somewhere else. Ending with what would settle each open question lets people judge the next update.

TAKEAWAY 19W

A public account is ordered by the strength of its evidence, not by the order the work happened in.

10.3 What survives the emergency

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · CHOICE

SCENE 25W

The emergency budget ends in three weeks, and anything not protected stops with it. Four programmes want the last line. The council votes this month.

GUIDE 57W

Four programmes, and three of them worked this time. Ask of each whether it would work on a different outbreak. Each stream sees a different part, and the value comes from putting them side by side. Note what the cheapest thing on the list costs too. Records are nearly free, and they are the first to go.

ASK 11W

Which capability should the city protect when the emergency budget closes?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

anything not funded to continue stops when the emergency does

VERDICT 80W

Each stream sees a different part of an outbreak, and each is much less useful alone. Sequencing says what changed, and it needs samples that stand for the town. Sewage gives the earliest warning, and it names nobody and explains nothing. Hospital numbers say who is ill, and they arrive late. What made this response work was the three being read together. That is the thing that quietly disappears when the money ends, because no single line item owns it.

TAKEAWAY 16W

Keep the system that spots the next one, not the instrument that starred in this one.

10.4 What the marquee is taking now

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · CALLBACK · SPOT

SCENE 25W

People arrive through the marquee all afternoon. One rule on the whiteboard says who is seen first, and it is rewritten as the ward fills.

GUIDE 33W

Sort arrivals by the rule on the board and leave the rest. The rule changes during the afternoon and nobody calls it out. What counts is the arrivals either side of a change.

ASK 12W

Sort arrivals to the rule on the whiteboard, and keep checking it.

BEHIND THE BUTTON 60W

Why the rule changes. It starts with how ill somebody is. When the isolation beds fill, it becomes who cannot safely wait at home. When the swabs come back, it becomes who is confirmed.

Why the change is where harm happens. Ten minutes on the old rule sends confirmed patients into the queue the new rule was written to clear.

TAKES AS READ

a sorting rule can be changed while people are still arriving

VERDICT 111W

Three rules run across the afternoon and each wants different people. Anybody short of breath, then anybody who cannot stay away from others at home, then anybody with a confirmed test. A person can match two at once, and that is what makes a change cost something real. What the panel scores is the arrivals either side of a change. Most people who come through are wanted by neither rule, so handling those correctly proves nothing. The marquee's own version is the quarter hour after the isolation beds fill. The rule has changed, and the people already being walked down the corridor are being walked there under the rule from before.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

END OF THE STAGE 19W

Use evidence synthesis, causal chains, uncertainty, ethics, communication, long-term monitoring as an evidence chain rather than as isolated facts.

THE CLOSING CARD

Forty-two days with no new case, and the outbreak was declared over on a Tuesday afternoon. The treatment works. The case definition held from the third week to the last. The animal reservoir was settled in the end — not by the briefing, but by the survey the briefing said was still needed.

The clinics stayed open, the surveillance stayed funded, and the curve that made the city frightened in March is now a figure in a report somebody else will learn from.

Card text only — no options, boards or answers. 7,485 words of card prose across 40 stops. Generated from `themes/outbreak_riverton_ms` by `tools/make-cardbook.mjs`.